

STAT 4714 — Final Exam Study Guide (Solutions + Flash Cards)

Max Miller

2026-05-10

STAT 4714 — Final Exam Study

Flash Cards (2 index cards, front + back, 11 lines per side)

Note: PMF/PDF formulas and mean/variance for **Bin**, **Geom**, **NegBin**, **Hypergeom**, **Pois**, **Normal**, **Exp**, **Gamma**, **Weibull** are on the formula sheet — cards focus on what is NOT there. The sheet also allows shorthand answers like $b(X = 2; n = 12, p = 0.8)$.

Card 1 — Front: Probability rules, distribution ID, key relationships

1. **Bayes:** $P(A | B) = \frac{P(B | A)P(A)}{P(B)}$. **Total prob:** $P(B) = \sum_i P(B | A_i)P(A_i)$.
2. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. **Indep:** $P(A \cap B) = P(A)P(B)$. **Cond:** $P(A | B) = P(A \cap B)/P(B)$.
3. **ID dists:** fixed n trials count successes $\rightarrow$ **Bin**; til 1st succ $\rightarrow$ **Geom**; til k th succ $\rightarrow$ **NegBin**; rate over interval/area $\rightarrow$ **Pois**; sample w/o replacement $\rightarrow$ **Hypergeom**.
4. **Series rel** = $\prod R_i$ (need ALL). **Parallel rel** = $1 - \prod(1 - R_i)$ (need ANY). Sys fails iff every path fails.
5. **Pois** $\square$ **Exp** $\square$ **Gamma:** $\text{Pois}(\lambda t)$ counts events; $\text{Exp}(\beta = 1/\lambda)$ = wait til NEXT; $\text{Gamma}(\alpha, \beta = 1/\lambda)$ = wait til α th. Sum of α iid $\text{Exp}(\beta)$ = $\text{Gamma}(\alpha, \beta)$.
6. **Memoryless** (Exp & Geom ONLY): $P(X > s+t | X > s) = P(X > t)$. Uniform/Normal/Gamma($\alpha > 1$) NOT memoryless.
7. **Norm approx Bin:** need $np \geq 5$ AND $n(1-p) \geq 5$. CC: $P(X \leq a) \approx \Phi(\frac{a+0.5-np}{\sqrt{np(1-p)}})$; $P(X \geq a) \approx 1 - \Phi(\frac{a-0.5-np}{\sqrt{np(1-p)}})$.
8. **Pois approx Bin:** n large, p small $\rightarrow$ $\text{Pois}(\lambda = np)$.
9. $Z = (X - \mu)/\sigma$. 90%: ± 1.645 , 95%: ± 1.96 , 99%: ± 2.576 .
10. **Chebyshev:** $P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$ (or $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$). Distribution-free LOWER bound. **Empirical (normal):** 68/95/99.7 within $1/2/3\sigma$.
11. **Gamma–Poisson tail (integer α):** $P(T > t) = \sum_{k=0}^{\alpha-1} \frac{e^{-t/\beta}(t/\beta)^k}{k!}$.

Card 1 — Back: CDF/quantile, joint distributions, covariance

1. Continuous: $f \geq 0$, $\int f = 1$. $F(x) = \int_{-\infty}^x f$. $P(a < X < b) = F(b) - F(a)$. $f = F'$.
2. **CDF** $\Phi(z)$: value $\rightarrow$ prob. **Quantile** $\Phi^{-1}(p)$: prob $\rightarrow$ value. Inverses. CI uses $\Phi^{-1}(\alpha/2)$, $\Phi^{-1}(1-\alpha/2)$.

- Joint cont: $P((X, Y) \in A) = \iint_A f$. **Marginal** $f_X(x) = \int f(x, y) dy$, $f_Y(y) = \int f(x, y) dx$.
- Conditional** $f(y | x) = f(x, y) / f_X(x)$. Then $P(a < Y < b | X = x) = \int_a^b f(y | x) dy$.
- Independence** $\Leftrightarrow f(x, y) = f_X(x) f_Y(y)$ for ALL (x, y) AND support is rectangular (e.g., $0 < x < y < 1$ is NOT rect).
- $E[g(X, Y)] = \iint g(x, y) f(x, y)$. $E[XY] = \iint xy f$.
- Cov**: $\sigma_{XY} = E[XY] - \mu_X \mu_Y$. **Indep** $\Rightarrow \sigma_{XY} = 0$; **Cov** $= 0 \nRightarrow$ indep (could be nonlinear).
- Correlation**: $\rho_{XY} = \sigma_{XY} / (\sigma_X \sigma_Y)$, $\rho \in [-1, 1]$. Scale-free.
- $\text{Var}(aX + bY + c) = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \sigma_{XY}$. **Indep**: drop the cross term.
- CDF $\rightarrow$ PDF: differentiate. PDF $\rightarrow$ prob: integrate over interval. PDF $\rightarrow$ CDF: integrate from $-\infty$.
- Survival: $P(T > t) = 1 - F(t)$. For $\text{Exp}(\beta)$: $P(T > t) = e^{-t/\beta}$.

Card 2 — Front: Sampling, CI, hypothesis testing, bootstrap

- $\text{SE}(\bar{X}) = \sigma / \sqrt{n}$ (use $s / \sqrt{n}$ if σ unknown). **CLT**: $\bar{X} \stackrel{\text{approx}}{\sim} N(\mu, \sigma^2/n)$ for large n , ANY pop.
- CI = estimate $\pm$ crit $\cdot$ SE. Unknown σ , normal pop: $\bar{X} \pm t_{\alpha/2, n-1} \cdot s / \sqrt{n}$.
- SE** = SD of estimator (precision, scale). **CI** = plausible param range; can be asymmetric (esp. bootstrap).
- p-value** = $P(\text{stat as/more extreme} | H_0)$. Upper H_a : right tail. Lower: left. Two-sided: 2-tail or $P(|T| \geq |t_{\text{obs}}|)$.
- Type I** = reject true H_0 (rate α). **Type II** = retain false H_0 (rate β). **Power** = $1 - \beta$.
- $\uparrow n$: $\beta \downarrow$, power $\uparrow$ (no Type-I cost). $\uparrow \alpha$: power $\uparrow$ but more false alarms. Larger effect $\rightarrow$ power $\uparrow$.
- Decision**: $p < \alpha$ reject H_0 ; else fail to reject. Confidence = $1 - \alpha$ (90% CI $\leftrightarrow \alpha = 0.10$).
- Bootstrap (CI for any $\hat{\theta}$)**: (1) resample n WITH replacement; (2) compute $\hat{\theta}^*$; (3) repeat $B \geq 1000$; (4) percentile CI = $[\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*]$; SE = $\text{sd}(\hat{\theta}^*)$.
- Parametric sim test**: (1) sim B samples from H_0 model; (2) compute stat each; (3) $p \approx \text{frac as/more extreme than obs}$. Sim under H_0 for null reference dist.
- Bootstrap test**: re-center sample so H_0 holds, then bootstrap stats and compute p .
- Sim **pros**: no parametric assumption, weird stats OK. **Cons**: random/compute cost; need credible null (misspecified $\rightarrow$ bad p).

Card 2 — Back: Simple Linear Regression (Ch 11)

- Model: $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $\varepsilon_i \text{ iid } N(0, \sigma^2)$. Assumes **Linearity**, **Independence**, **const Variance**, **Normality** of errors.
- LSE** minimize $SSE = \sum (y_i - \hat{y}_i)^2$: $b_1 = \frac{S_{xy}}{S_{xx}}$, $b_0 = \bar{y} - b_1 \bar{x}$ where $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$, $S_{xx} = \sum (x_i - \bar{x})^2$.
- $SST = \sum (y_i - \bar{y})^2$ total. $SSE = \sum (y_i - \hat{y}_i)^2$ unexplained. $SSR = SST - SSE = \sum (\hat{y}_i - \bar{y})^2$ explained.
- $R^2 = SSR / SST = 1 - SSE / SST$ = frac variance explained. HIGH $R^2 \neq$ correct linear model — CHECK RESIDUALS.
- CI for mean $\mu_{Y|x_0}$** : narrow, avg behavior. **PI for new y_0** : WIDER (adds ε noise on top of fit uncertainty). $\text{PI} \supset \text{CI}$.
- Bootstrap CI mean@ x_0** : (1) fit, get $\hat{y}_i$ + residuals e_i ; (2) resample residuals, build $y^* = \hat{y} + e^*$, refit b_0^*, b_1^* , compute $b_0^* + b_1^* x_0$; (3) percentile of bootstrap dist.
- Bootstrap PI**: $y_0^* = b_0^* + b_1^* x_0 + e^*$ (add random residual for individual noise); percentile interval.

8. **Randomization** $H_0 : \beta_1 = 0$: SHUFFLE y vs x (break link); refit b_1^* many times; $p = \text{frac } |b_1^*| \geq |b_{1,\text{obs}}|$. Nonparametric.
9. **Residuals** $e_i = y_i - \hat{y}_i$. Plot e_i vs $\hat{y}_i$ (or x_i): want random scatter ~ 0 , no pattern.
10. **Diagnostics**: fan/funnel $\rightarrow$ heteroscedasticity (var grows w/ $\hat{y}$). Curve $\rightarrow$ nonlinear. **QQ-plot** off straight line / heavy tails $\rightarrow$ non-normal ε .
11. **NEVER extrapolate** beyond observed x range. Significant $\hat{\beta}_1$ + bad residuals $\Rightarrow$ SE/PI unreliable. R^2 doesn't validate the model form.

Worked Solutions (Q1–Q24)

Question 1

Let U = upper path works, L = lower path works.

(a) By the **series rule** $R_{\text{series}} = \prod_i R_i$:

$$P(U) = (0.85)(0.90) = 0.765, \quad P(L) = (0.80)(0.80)(0.95) = 0.608.$$

By the **parallel rule** $R_{\text{parallel}} = 1 - \prod_i (1 - R_i)$:

$$P(\text{works}) = 1 - (1 - 0.765)(1 - 0.608) = 1 - (0.235)(0.392) = \boxed{0.90788}.$$

(b) Given the system works and the upper path failed, the lower path must work:

$$P(U^c \cap \text{works}) = P(U^c \cap L) = P(U^c)P(L) = (0.235)(0.608) = 0.14288,$$

$$P(U^c \mid \text{works}) = \frac{0.14288}{0.90788} = \boxed{0.15738}.$$

(c) The system fails iff **both** paths fail, so $\{\text{system fails}\} = U^c \cap L^c \subseteq U^c$:

$$P(U^c \mid \text{fails}) = \frac{P(U^c \cap L^c)}{P(U^c \cap L^c)} = \boxed{1}.$$

Question 2

Let F = faulty, L = labeled “faulty.” Given $P(F) = 0.04$, $P(L \mid F) = 0.90$, $P(L \mid F^c) = 0.07$.

(a) Law of total probability:

$$P(L) = (0.90)(0.04) + (0.07)(0.96) = 0.036 + 0.0672 = \boxed{0.1032}.$$

(b) Bayes’ theorem:

$$P(F \mid L) = \frac{P(L \mid F)P(F)}{P(L)} = \frac{0.036}{0.1032} \approx \boxed{0.3488}.$$

Question 3

(a) Antiderivative: $\int 3(1-x)^2 dx = -(1-x)^3 + C$.

$$P(X > 0.6) = \left[-(1-x)^3 \right]_{0.6}^1 = (0.4)^3 = \boxed{0.064}.$$

(b)

$$P(0.6 < X < 0.8) = \left[-(1-x)^3 \right]_{0.6}^{0.8} = -(0.2)^3 + (0.4)^3 = 0.056.$$

$$P(X < 0.8 \mid X > 0.6) = \frac{0.056}{0.064} = \boxed{\frac{7}{8} = 0.875}.$$

Question 4

(a) Marginals:

$$f_X(x) = \int_0^\infty 4e^{-2x-2y} dy = 2e^{-2x}, \quad f_Y(y) = 2e^{-2y}.$$

Conditional at $y = 1$: $f(x | y = 1) = 2e^{-2x}$.

$$P(1 < X < 2 | Y = 1) = \int_1^2 2e^{-2x} dx = e^{-2} - e^{-4} \approx \boxed{0.1170}.$$

(b) Since $f_X(x)f_Y(y) = 4e^{-2x-2y} = f(x, y)$ for all $x, y > 0$, X and Y are independent.

(c) Independence implies $E[XY] = E[X]E[Y] = \mu_X\mu_Y$, hence $\sigma_{XY} = 0$.

Question 5

(a) $f(t) = F'(t) = \frac{1}{40}e^{-t/40}$, $t > 0$. This is $\text{Exp}(\beta = 40)$.

(b) $P(T > 90) = 1 - F(90) = e^{-90/40} = e^{-2.25} \approx \boxed{0.1054}$.

(c) $F(90) \approx 0.8946$ means about 89.5% of capacitors fail by time $t = 90$. Equivalently, the probability that a capacitor's lifetime ends at or before 90 time units is ≈ 0.8946 .

Question 6

Joint density: $f(x, y) = 6x$ on $0 < x < y < 1$.

Marginals:

$$f_X(x) = \int_x^1 6x dy = 6x(1 - x), \quad f_Y(y) = \int_0^y 6x dx = 3y^2.$$

Means: $\mu_X = \int_0^1 x \cdot 6x(1 - x) dx = \frac{1}{2}$, $\mu_Y = \int_0^1 y \cdot 3y^2 dy = \frac{3}{4}$.

Second moments: $E[X^2] = \frac{3}{10}$, $E[Y^2] = \frac{3}{5}$.

Variances: $\sigma_X^2 = \frac{3}{10} - \frac{1}{4} = \frac{1}{20}$, $\sigma_Y^2 = \frac{3}{5} - \frac{9}{16} = \frac{3}{80}$.

Cross moment:

$$E[XY] = \int_0^1 \int_x^1 xy \cdot 6x dy dx = \int_0^1 6x^2 \cdot \frac{1-x^2}{2} dx = \frac{2}{5}.$$

Covariance: $\sigma_{XY} = \frac{2}{5} - \frac{1}{2} \cdot \frac{3}{4} = \frac{1}{40}$.

Correlation: $\sigma_X\sigma_Y = \sqrt{\frac{1}{20} \cdot \frac{3}{80}} = \frac{\sqrt{3}}{40}$, so

$$\rho_{XY} = \frac{1/40}{\sqrt{3}/40} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \approx 0.5774.$$

Question 7

$\mu = 1000, \sigma = 80$. Half-width 180, so $k = 180/80 = 2.25$.

Apply Chebyshev:

$$P(820 < T < 1180) \geq 1 - \frac{1}{(2.25)^2} = 1 - \frac{1}{5.0625} \approx \boxed{0.8025}.$$

Comment: This is a distribution-free lower bound. If T were known to be normal, the actual probability would be ≈ 0.976 . Chebyshev is conservative — useful when only μ and σ are known.

Question 8

(a) $X \sim \text{Bin}(4, 0.9)$.

$$P(X = 0) = (0.1)^4 = 0.0001, \quad P(X = 1) = 4(0.9)(0.1)^3 = 0.0036.$$

$$P(X \geq 2) = 1 - 0.0001 - 0.0036 = \boxed{0.9963}.$$

(b) Since $0.9963 \geq 0.99$, the 4-sensor system **already meets** the reliability target.

For comparison, majority-rule with $n = 3$ (need $X \geq 2$):

$$P = \binom{3}{2} (0.9)^2 (0.1) + (0.9)^3 = 0.243 + 0.729 = 0.972 < 0.99.$$

For $n = 5$ (need $X \geq 3$):

$$P = \binom{5}{3} (0.9)^3 (0.1)^2 + \binom{5}{4} (0.9)^4 (0.1) + (0.9)^5 = 0.99144 \geq 0.99.$$

(c) Positive correlation makes sensor failures **cluster** — when one fails, others are more likely to fail simultaneously. This violates the independence assumption, increases the variance of X , and pushes more probability mass into the lower tail (more multi-sensor failures). System reliability **decreases** below 0.9963.

Question 9

(a) **Binomial**(n, p): fixed number n of independent Bernoulli trials with constant p , counting successes.

(b) **Geometric**(p), with $p = P(\text{tail})$: independent Bernoulli trials repeated until the first success; X = trial number of that first success.

(c) **Poisson**(λ), with λ = expected packets per 30 s: counts of memoryless, independent arrivals in a fixed interval.

(d) **Negative Binomial**($k = 3, p$), with $p = P(\text{failure on a single test})$: trials repeated until a fixed number $k = 3$ of “successes” (failures) occurs; X = total trials needed.

Question 10

$X \sim \text{Bin}(35, 0.08), Y \sim \text{NegBin}(k = 4, p = 0.08)$.

(a) $\binom{35}{4} = 52,360$, $(0.08)^4 = 4.096 \times 10^{-5}$, $(0.92)^{31} \approx 0.07720$.

$$P(X = 4) = 52,360 \cdot (4.096 \times 10^{-5}) \cdot 0.07720 \approx \boxed{0.1656}.$$

(b) $\binom{34}{3} = 5,984$.

$$P(Y = 35) = 5,984 \cdot (0.08)^4 \cdot (0.92)^{31} \approx \boxed{0.01893}.$$

(Equivalently: $P(Y = 35) = \frac{4}{35}P(X = 4)$.)

(c) In (a), X is **binomial**: $n = 35$ trials is **fixed**, count failures (random). In (b), Y is **negative binomial**: $k = 4$ failures is **fixed**, count trials needed (random). Critically, (b) requires the 4th failure to land *exactly* on trial 35.

Question 11

(a) $X \sim \text{Bin}(300, 0.03)$. $\mu = 9$, $\sigma^2 = 8.73$, $\sigma \approx 2.955$.

Continuity correction: $P(X \leq 12) \approx P(X \leq 12.5)$.

$$P(X \leq 12) \approx \Phi\left(\frac{12.5 - 9}{2.955}\right) = \Phi(1.184) \approx \boxed{0.882}.$$

(b) Conditions: $np \geq 5$ and $n(1 - p) \geq 5$. Verify: $np = 9 \geq 5$ $\square$, $n(1 - p) = 291 \geq 5$ $\square$. Both conditions met. (Since p is small, a Poisson approximation with $\lambda = 9$ would also work.)

Question 12

(a) A Poisson process at rate λ produces $N(t) \sim \text{Poisson}(\lambda t)$ events in interval of length t . Waiting time T until next event satisfies $P(T > t) = P(N(t) = 0) = e^{-\lambda t}$, so $T \sim \text{Exp}(\beta = 1/\lambda)$. Counts and inter-arrival times are two views of the same process.

(b) Wait time until the α th event = sum of α iid $\text{Exp}(1/\lambda)$ inter-arrivals = $\text{Gamma}(\alpha, \beta = 1/\lambda)$. Shape α = how many events we are waiting for; $\alpha = 1$ recovers the exponential.

(c) Call center receiving calls at $\lambda = 5/\text{min}$: number of calls in 10 min $\sim \text{Poisson}(50)$; time to first call $\sim \text{Exp}(\beta = 1/5)$; time to second call $\sim \text{Gamma}(\alpha = 2, \beta = 1/5)$.

Question 13

(a) Since $X \sim \text{Uniform}[0, 10]$, $P(X > a) = (10 - a)/10$:

$$P(X > 9 \mid X > 7) = \frac{P(X > 9)}{P(X > 7)} = \frac{1/10}{3/10} = \boxed{\frac{1}{3}}.$$

(b) Unconditionally $P(X > 9) = 0.1 \neq 1/3$. Conditioning on $\{X > 7\}$ restricts the sample space to $[7, 10]$, so $X \mid X > 7 \sim \text{Uniform}[7, 10]$ — a smaller window where the chance of exceeding 9 is larger. The uniform is **not memoryless**: $P(X > s + t \mid X > s) \neq P(X > t)$. Only the exponential (among continuous distributions) has this property.

Question 14

Rate $\lambda = 8/20 = 0.4$ defects/m. For 5 m: $\lambda t = 2$, so $X \sim \text{Poisson}(2)$.

$$P(X < 2) = P(X = 0) + P(X = 1) = e^{-2} + 2e^{-2} = 3e^{-2} \approx \boxed{0.4060}.$$

Question 15

Step 1: $T \sim \text{Exp}(\beta = 3)$, so $p = P(T \leq 1) = 1 - e^{-1/3} \approx 0.2835$.

Step 2: $Y = \# \text{ failures} \sim \text{Bin}(120, 0.2835)$. $\mu = np \approx 34.02$, $\sigma^2 \approx 24.38$, $\sigma \approx 4.94$.

Step 3: Conditions met ($np, n(1-p) \geq 5$). Normal approximation with CC:

$$P(Y \leq 30) \approx \Phi\left(\frac{30.5 - 34.02}{4.94}\right) = \Phi(-0.713) \approx \boxed{0.2379}.$$

Question 16

(a) $E(T) = \alpha\beta = 5 \cdot 10 = \boxed{50 \text{ weeks}}$.

(b) Since $\alpha = 5$ is integer, use the gamma–Poisson relationship with $30/\beta = 3$:

$$P(T > 30) = \sum_{k=0}^4 \frac{e^{-3} 3^k}{k!} = e^{-3}(1 + 3 + 4.5 + 4.5 + 3.375) = e^{-3}(16.375) \approx \boxed{0.8153}.$$

Question 17

(a) The **CDF** $\Phi(z)$ takes a value z and outputs a probability $P(Z \leq z)$. The **quantile function** $\Phi^{-1}(p)$ takes a probability p and outputs the value z where $P(Z \leq z) = p$. They are inverses.

(b) **CDF use:** computing tail probability $P(Z \leq 1.5) = \Phi(1.5) \approx 0.9332$. **Quantile use:** finding critical value, e.g., $\Phi^{-1}(0.95) \approx 1.645$.

(c) $\boxed{[\Phi^{-1}(0.025), \Phi^{-1}(0.975)] = [z_{0.025}, z_{0.975}] = [-1.96, 1.96]}$.

Question 18

(a) $X \sim \text{Poisson}(\mu = \lambda t = 4 \cdot 0.75 = 3)$ runners in 45 min.

(b) $T \sim \text{Exp}$ with rate $\lambda = 4/\text{hr}$, mean $\beta = 1/\lambda = 0.25 \text{ hr}$ (15 min).

(c) $P(T > 0.5) = e^{-\lambda t} = e^{-4 \cdot 0.5} = e^{-2} \approx \boxed{0.1353}$.

Question 19

(a) **p-value depends on H_a :** The p -value is $P(\text{stat as/more extreme} \mid H_0)$, but “extreme” is defined by which direction H_a points. For test stat T with observed t_{obs} : **upper-tailed** ($\theta > \theta_0$) gives $p = P(T \geq t_{\text{obs}} \mid H_0)$ (right tail); **lower-tailed** gives left tail; **two-sided** counts both tails (e.g., $p = 2P(T \geq |t_{\text{obs}}| \mid H_0)$). Same t_{obs} , different p .

(b) **SE vs CI:** **SE** is a scalar measure of estimator precision — useful for comparing estimators or propagating uncertainty. **CI** translates SE into a calibrated plausible range for θ at confidence $1 - \alpha$, communicating asymmetry too (a percentile bootstrap CI need not be symmetric around $\hat{\theta}$). SE alone hides asymmetry and the confidence level; CI alone obscures the underlying scale of variability.

(c) **Bootstrap procedure:** Given data $\{x_1, \dots, x_n\}$ and $\hat{\theta} = T(x_1, \dots, x_n)$: 1. **Resample** with replacement to get $\{x_1^*, \dots, x_n^*\}$ of size n . 2. **Compute** $\hat{\theta}^{*(b)} = T(x_1^*, \dots, x_n^*)$. 3. **Repeat** for $b = 1, \dots, B$ ($B \geq 1000$). 4. **Summarize:** histogram, $\widehat{\text{SE}} = \text{sd}(\hat{\theta}^*)$, percentile CI $[\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*]$.

(d) Type I, II, Power: **Type I** = reject true H_0 (rate α). **Type II** = retain false H_0 (rate β). **Power** = $1 - \beta$ = chance of correctly rejecting. **Increasing** α makes rejection easier — power $\uparrow$ but more false alarms. **Increasing** n tightens both null and alternative distributions, reducing overlap, so $\beta \downarrow$ and power $\uparrow$ *without* inflating α — the preferred lever.

Question 20

Approach (i): Parametric (CLT-based): 1. Collect n independent measurements $x_1, \dots, x_n$. 2. Compute $\bar{X} = \frac{1}{n} \sum x_i$ and $s^2 = \frac{1}{n-1} \sum (x_i - \bar{X})^2$. 3. Estimate $\widehat{SE}(\bar{X}) = s/\sqrt{n}$. 4. By CLT, $\bar{X} \overset{\text{approx}}{\sim} \mathcal{N}(\mu, s^2/n)$ — use a t_{n-1} reference for small n . 5. Form $(1 - \alpha)$ CI: $\bar{X} \pm t_{\alpha/2, n-1} \cdot s/\sqrt{n}$.

Assumptions: iid measurements with finite variance; n large for CLT (or population approximately normal so t -approx holds); no heavy tails / extreme outliers.

Approach (ii): Bootstrap (resampling): 1. Treat the observed sample as a stand-in for the population. 2. Draw $\{x_1^*, \dots, x_n^*\}$ of size n from the sample **with replacement**. 3. Compute $\bar{X}^* = \frac{1}{n} \sum x_i^*$. 4. Repeat B times (e.g., $B = 5000$) to get $\bar{X}_1^*, \dots, \bar{X}_B^*$. 5. Histogram = bootstrap sampling distribution. $\widehat{SE} = \text{sd}(\bar{X}^*)$. Percentile CI from quantiles.

Assumptions: sample is representative (unbiased random sample); observations independent; n large enough that the empirical distribution adequately approximates the population.

Question 21

(a) Parametric simulation procedure: 1. Under H_0 , simulate B (e.g., $B = 10,000$) samples of size $n = 25$ from $N(4.8, 0.6^2)$. 2. For each simulated sample j , compute $\bar{X}_j^*$. 3. Compute two-sided distance $D_j = |\bar{X}_j^* - 4.8|$. 4. Observed distance: $D_{\text{obs}} = |5.05 - 4.8| = 0.25$. 5. Approximate $p \approx \frac{1}{B} \sum \mathbf{1}\{D_j \geq 0.25\}$.

(b) Why null model: The p -value asks how unusual the observed data would be if H_0 were true. The only way to assess that is to know the sampling distribution of the test statistic under H_0 — simulating under H_0 generates that null reference distribution empirically.

(c) Decision ($\alpha = 0.10$): $p \approx 0.062 < 0.10$, so REJECT H_0 . Sufficient evidence at 90% confidence that $\mu \neq 4.8$ V.

(d) Pro: No closed-form parametric assumption about the test statistic — works for unusual or analytically intractable statistics. **Con:** Monte Carlo / computational cost and randomness in the p -value; you still need a credible null model — a misspecified null gives a misleading p .

Question 22

(a) SST/SSR/SSE/ R^2 : $SST = \sum (y_i - \bar{y})^2$ is total variability in y about its mean. $SSE = \sum (y_i - \hat{y}_i)^2$ is unexplained (residual) variability. $SSR = SST - SSE = \sum (\hat{y}_i - \bar{y})^2$ is variability explained by the model. $R^2 = SSR/SST = 1 - SSE/SST$ is the proportion of total variability the model explains; $R^2 = 1$ is perfect fit, $R^2 = 0$ no better than $\bar{y}$.

(b) Randomization test for $H_0 : \beta_1 = 0$: *Logic:* Under H_0 , x and y are unrelated, so any pairing is equally plausible. Shuffling y vs x destroys real linear link and builds the chance distribution of b_1 . *Steps:* (1) Fit SLR on original data, record $b_{1, \text{obs}}$. (2) For $b = 1, \dots, B$: permute y against fixed x , refit, store $b_1^{*(b)}$. (3) Two-sided $p = \frac{1}{B} \sum \mathbf{1}\{|b_1^{*(b)}| \geq |b_{1, \text{obs}}|\}$. (4) Reject if $p < \alpha$.

(c) Least squares: Minimize $SSE = \sum (y_i - b_0 - b_1 x_i)^2$. Setting partials to zero gives the normal equations, which solve to:

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad b_0 = \bar{y} - b_1 \bar{x}.$$

(d) CI vs PI: CI for mean response $\mu_{Y|x_0}$ captures uncertainty in the estimated regression *line* at x_0 . **PI for new** y_0 captures that line uncertainty **plus** the individual error $\varepsilon \sim N(0, \sigma^2)$. PI is **wider** — variance is roughly $\widehat{\text{Var}}(\hat{\mu}_{x_0}) + \hat{\sigma}^2$ vs just $\widehat{\text{Var}}(\hat{\mu}_{x_0})$.

(e) CI for mean response @ x_0 (residual bootstrap): (1) Fit SLR, get $\hat{y}_i$ and residuals $e_i = y_i - \hat{y}_i$. (2) For $b = 1, \dots, B$: resample residuals with replacement to get e_i^* , form $y_i^* = \hat{y}_i + e_i^*$, refit to get b_0^*, b_1^* , compute $\hat{\mu}_{x_0}^* = b_0^* + b_1^* x_0$. (3) 95% percentile CI = $(Q_{0.025}(\hat{\mu}^*), Q_{0.975}(\hat{\mu}^*))$. Quantities estimated: b_0, b_1 , residuals $\{e_i\}$, and the bootstrap distribution of $\hat{\mu}_{x_0}^*$.

(f) Residual analysis: (i) *Nonconstant variance:* Plot e_i vs $\hat{y}_i$. Under homoscedasticity, points form a horizontal band of constant width around 0. **Fan/funnel** shape (spread grows or shrinks with $\hat{y}$) indicates heteroscedasticity. Fix: variance-stabilizing transform (e.g., $\log y$, $\sqrt{y}$). (ii) *Normality:* Make a **Q-Q plot** of residuals vs theoretical normal quantiles; points should fall on the 45° line. Systematic curvature, S-shapes, or heavy tails (points peeling away at the ends) signal non-normality. A residual histogram should also look symmetric and bell-shaped.

Question 23

(a) Flaw in Claim 1: A high R^2 (0.88) only quantifies variance explained — it does NOT validate that linearity is the correct structural form. R^2 is blind to systematic curvature, heteroscedasticity, influential outliers, and correlated errors; a nonlinear model could fit the same data substantially better while the linear R^2 is still high. Residual plots are essential to diagnose model adequacy and cannot be skipped.

(b) Flaw in Claim 2: This is **extrapolation**. The model was fit on $x \in (20^\circ\text{C}, 80^\circ\text{C})$, so predictions at -150°C lie far outside the training range where the linear relationship has no empirical support. The true relationship may be nonlinear or break down entirely outside the window — and mechanically, $\hat{Y} = 1500 - 12(-150) = 3300$ hours, which is physically suspect for a temperature regime never observed.

Question 24

(a) Two violations: 1. **Constant variance (homoscedasticity):** the funnel/fan shape in residuals vs fitted is direct evidence of heteroscedasticity — $\text{Var}(\varepsilon_i)$ grows with x_i rather than equaling σ^2 . 2. **Independence of errors:** latency recorded every minute for 48 hours is time-series data, so consecutive residuals are temporally autocorrelated rather than independent.

(b) Compromises to PI: The PI formula uses a single pooled $\hat{\sigma}^2$ and assumes independent errors, so both violations distort its width. - *Heteroscedasticity:* $\hat{\sigma}^2$ averages low- and high-traffic variability — the 95% PI for the upcoming high-traffic event is **too narrow** (true variance at high x is much larger). Stated 95% coverage is not achieved. - *Autocorrelation:* positively correlated residuals make $\hat{\sigma}^2$ appear smaller than the true error variance, deflating SE and shrinking the PI further — overstating confidence.

Net effect: the admin's interval looks precise but systematically understates uncertainty exactly where prediction matters most.